Multi-messenger Astronomy

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Contents

The presentation will provide answers to the following questions:

- What is Multi-messenger Astronomy (MMA)?
- What are:
  - Cosmic Rays
  - Neutrinos
  - Gamma Rays
  - Gravitational Waves

  and what is so special about these four cosmic messengers?

What is Multi-messenger Astronomy?

Multi-messenger Astronomy (MMA) is the coordinated observation of cosmic phenomena using two or more distinct carriers of information,

- Electromagnetic radiation (radio to gamma rays)
- Gravitational waves
- Neutrinos
- Cosmic rays

Each messenger originates from different physical processes in the same source and reveals complementary information inaccessible by any single channel alone.

Why Multiple Messengers?

- Single-messenger observations give an incomplete picture of violent astrophysical events.
- Source parameters (mass, spin, composition, distance) are often degenerate when inferred from one signal alone.
- Cross-correlating independent messengers breaks these degeneracies.
- The power of MMA lies in time-coincident detection: a gravitational-wave chirp followed by a gamma-ray burst, a neutrino alert triggering an optical follow-up.

This reveals the full life-cycle of a violent event in a way no single telescope ever could.

The Four Cosmic Messengers

Cosmic Rays

- Charged particles (p, He, Fe)
- Energies up to $\sim 10^{20}$ eV
- Deflected by magnetic fields

Neutrinos

- Neutral, near-massless
- Weakly interacting
- Travel at $\approx c$, undeflected by fields
- Point directly back to sources

Gamma Rays

- Highest-energy EM photons (> 100 keV)
- Probe non-thermal processes
- Absorbed by EBL at TeV energies

Gravitational Waves

- Spacetime metric perturbations
- Generated by accelerating mass
- Travel at c, unobstructed

Cosmic Rays: The Highest-Energy Particles

Cosmic rays are charged particles (predominantly protons and nuclei) arriving from outside the Solar System.

The energy spectrum spans over 10 decades with notable spectral features,

- **The Knee** ($\sim 3 \times 10^{15}$ eV): spectral softening $E^{-2.7} \rightarrow E^{-3}$; marks end of efficient galactic confinement for light nuclei.
- **The Ankle** ($\sim 3 \times 10^{18}$ eV): spectral hardening; transition from galactic to extragalactic composition.
- **GZK suppression** ($\sim 5 \times 10^{19}$ eV): protons interact with CMB photons via $p + \gamma_{\mathrm{CMB}} \rightarrow \Delta^{+} \rightarrow p + \pi^{0}$, limiting propagation to $\lesssim 50$ Mpc.

Cosmic Rays: The Highest-Energy Particles

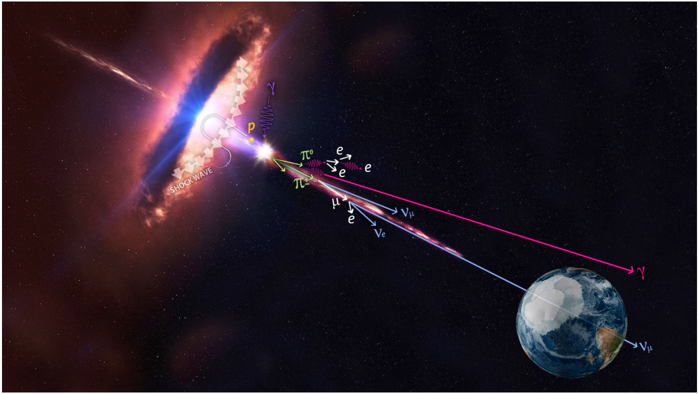

Figure: Cosmic Rays interacting with matter and radiation produces more highly energetic particles.

Cosmic Rays: The Highest-Energy Particles

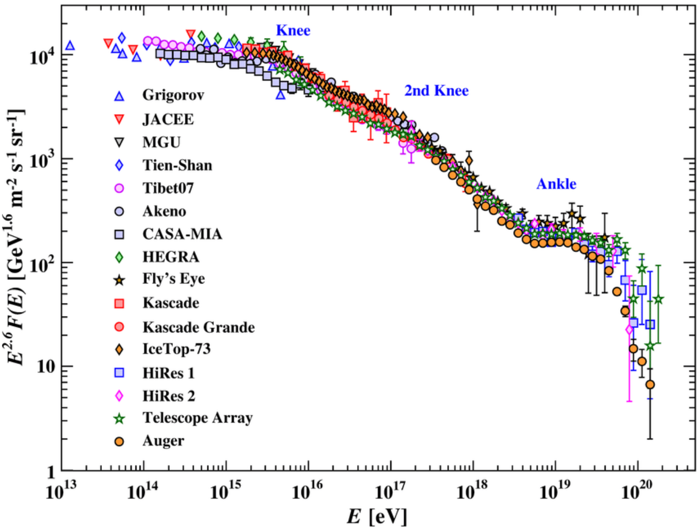

Figure: Cosmic Rays spectrum.

Cosmic Ray Acceleration

The standard mechanism is **diffusive shock acceleration**,

- Particles crossing a strong shock repeatedly gain.
- Naturally produces a power-law spectrum $N(E) \propto E^{-2}$.
- Maximum energy set by the Hillas criterion,

$$E_{\max} \approx ZeBR$$

  where Z is charge, B is the magnetic field, and R is the source size.

Auger Observatory results suggest a composition transition from light (proton-dominated) at the ankle to heavy (iron-dominated) near the GZK cutoff.

UHECR sources are co-producers of astrophysical neutrinos through pγ interactions.

Cosmic Ray Acceleration

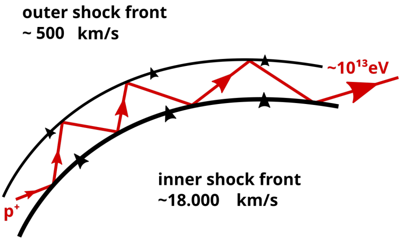

Figure: Illustration 1

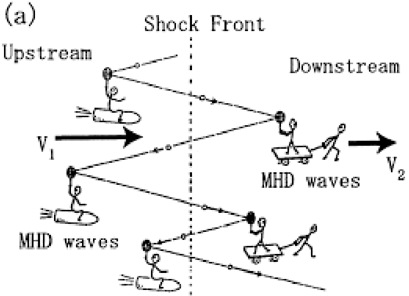

Figure: Illustration 2

Figure: Diffusive Shock Acceleration.

Cosmic Rays Detection

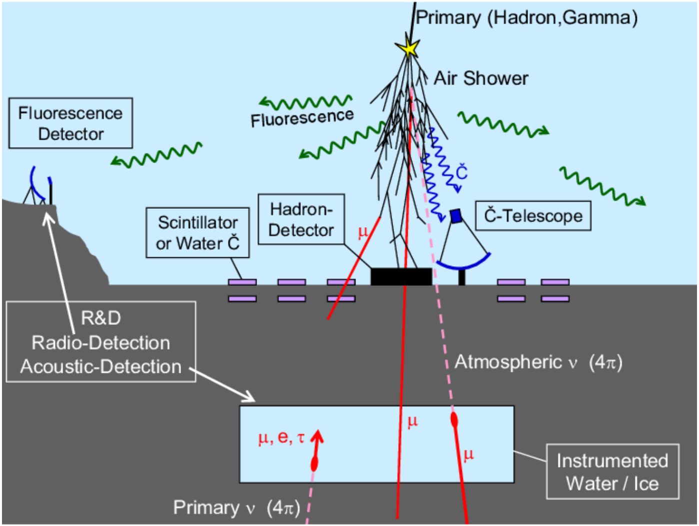

Figure: Cosmic Rays Detection.

Cosmic Ray Detection

The shower has three main components:

- **Electromagnetic (EM) component**: γ, e^+, e^- produced via pair production and bremsstrahlung. Carries the bulk of the shower energy.
- **Muonic component**: $\mu^\pm$ from charged pion and kaon decays ($\pi^\pm \to \mu^\pm + \nu_\mu$). Highly penetrating; reach the ground.
- **Hadronic component**: residual hadronic interactions producing $\pi^\pm$, π^0, K, etc. Acts as the energy reservoir feeding both other components.

Cosmic Ray Detection

There are several ways to detect Cosmic Rays on Earth, and they are used depending on which components we need to detect. Two of them are:

- **Water Cherenkov Detectors (WCDs)**: sensitive to both the EM and muonic components.
- **Scintillator detectors**: measure the EM component; cheaper but less sensitive to muons.

Neutrinos: The Ghost Particles

Cosmic-ray interactions with matter (pp, $p\gamma$) produce charged pions that subsequently decay,

$$\pi^+ \rightarrow \mu^+ + \nu_\mu, \quad \mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$$

Leading candidate sources of the diffuse astrophysical neutrino flux,

- Active Galactic Nuclei (AGN) jets
- Gamma-ray bursts (GRBs)
- Starburst galaxies

Even the Sun produces neutrinos of the order of 10^{38} per second. Earth receives a flux of the order of 10^{11} per cm^2 per second.

The IceCube Discovery

IceCube (South Pole, 1 km^3 instrumented ice) reported the first high-energy astrophysical neutrinos in 2013 (the "Bert and Ernie" events at ~ 1 PeV).

The diffuse astrophysical flux follows a power-law

$$\frac{d\Phi}{dE} \propto E^{-\gamma}, \quad \gamma \approx 2.0\text{–}2.5$$

consistent with Fermi-accelerated sources.

In 2022, NGC 1068 (a nearby Seyfert galaxy) was identified as a statistically significant point source at $\sim 4.2\sigma$.

Neutrino Detection

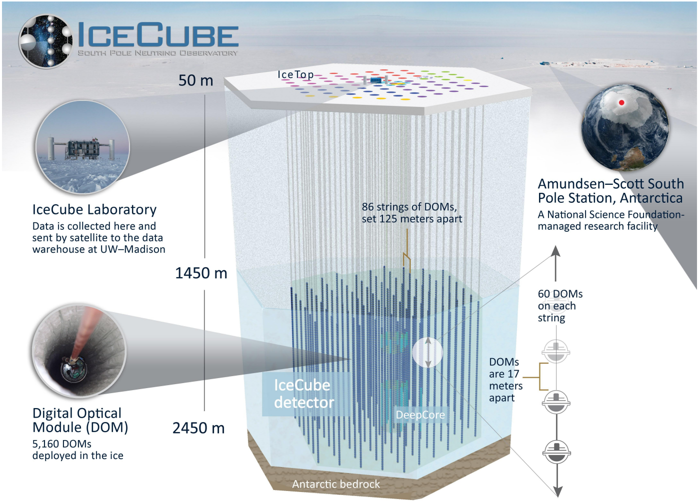

Figure: Neutrino Detection.

Neutrino Detection

Neutrinos interact only via the weak force and gravity. The cross-section (analogous to probability) of a collision is so less that a 1 MeV neutrino has a mean free path of $\sim 10^{18}$ cm in water, which is roughly 60 light years.

Consequences for detection:

- Detectors must be massive. Hundreds to thousands of tonnes of target material.
- Detectors must be shielded from cosmic-ray backgrounds. Deep underground, under ice, or under water.

For this, Water Cherenkov Detectors and Liquid Scintillator Detectors are used.

Gravitational Waves: Ripples in Spacetime

Predicted by Einstein's GR in 1916, gravitational waves are small disturbances in the fabric of space-time, travelling at the speed of light, c.

The strain is characterized by,

$$h = \frac{\Delta L}{L}$$

GW150914 (September 14, 2015): first direct detection by LIGO of two merging $\sim 30\,M_\odot$ black holes at ~ 1.3 Gpc, with peak strain $h \sim 10^{-21}$.

Gravitational Waves: Ripples in Spacetime

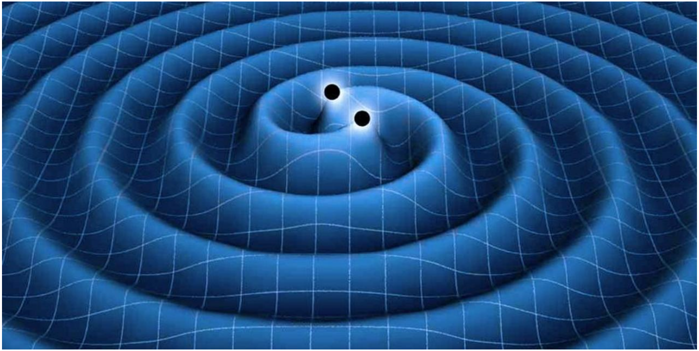

Figure: Gravitational Waves.

Gravitational Waves Detection

Figure: Illustration 1

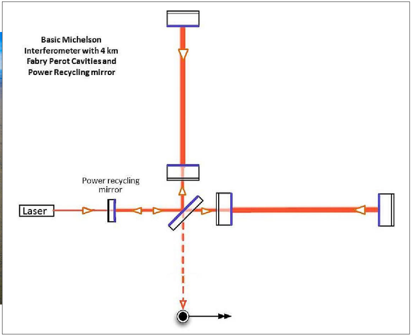

Figure: Illustration 2

Figure: Gravitational Waves Detection.

GW Source Classes

Theoretically, every accelerating mass emits gravitational waves but practically these ripples are so small that we can only detect it from extremely massive objects, like Black Holes and Neutron Stars, accelerating very quickly, like in a merger. This gives us the three classes of sources we can detect gravitational waves from,

- Binary Black Hole (BBH) mergers
- Binary Neutron Star (BNS) mergers
- Neutron Star–Black Hole (NSBH) mergers

Gamma Rays: The Most Energetic Light

Gamma rays ($E > 100$ keV) are produced by non-thermal, relativistic processes and serve as direct tracers of particle acceleration.

Understanding detector response requires knowing how gamma rays are produced:

- **Inverse Compton (IC)**: relativistic $e^{\pm}$ upscatter low-energy seed photons (CMB, optical, infrared).
- **Synchrotron self-Compton (SSC)**: the synchrotron photons emitted by the electrons themselves serve as the seed photon field for IC scattering.

Gamma Rays: The Most Energetic Light

- **Neutral pion decay** ($\pi^0 \to 2\gamma$): produced in hadronic pp or pγ interactions. Spectral pion bump at $\sim$ 70 MeV. Signature of hadronic acceleration in SNRs.
- **Bremsstrahlung**: $e^{\pm}$ deflected by ion Coulomb field; important in dense molecular clouds.
- **Curvature radiation / pair cascades**: in pulsar magnetospheres; γ-rays from magnetic pair cascades in the outer gaps.

Gamma Ray Emission Mechanisms

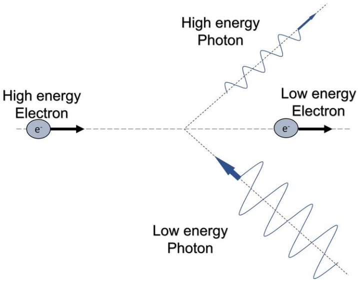

Figure: Illustration 1

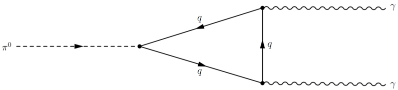

Figure: Illustration 2

Figure: Gamma Ray Emission Mechanisms.

Gamma Ray Emission Mechanisms

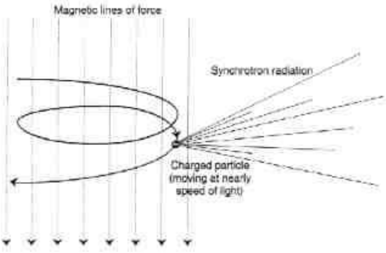

Figure: Illustration 3

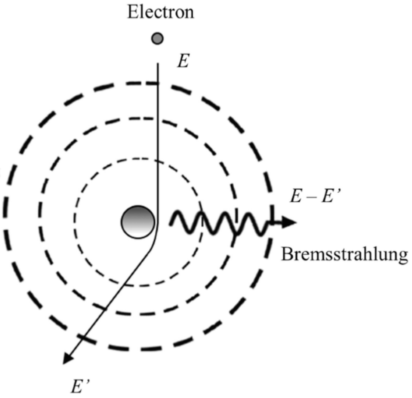

Figure: Illustration 4

Figure: Gamma Ray Emission Mechanisms.

Gamma Rays: The Most Energetic Light

Major source classes:

- **Gamma-Ray Bursts (GRBs)**: the most luminous transients in the Universe.
- **Active Galactic Nuclei**: blazars dominate the extragalactic γ-ray sky.
- **Pulsar Wind Nebulae**: Crab Nebula detected up to ~ 100 TeV via synchrotron and inverse Compton emission.
- **Supernova Remnants**: hadronic γ-rays from pion decay ($\pi^0 \to 2\gamma$). The pion bump at 70 MeV confirms CR acceleration in SNRs.

Gamma Rays: The Most Energetic Light

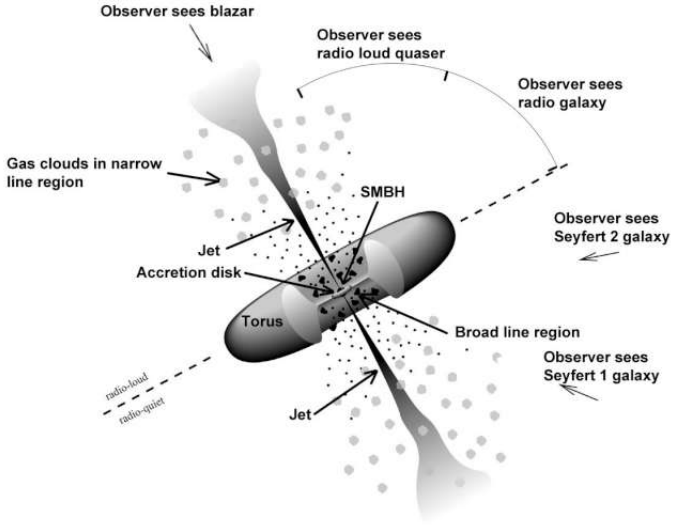

Figure: Active Galactic Nuclei.

Gamma Rays Detection

The two major detection techniques are:

- **Scintillation detectors**: an incoming photon deposits energy in a scintillating crystal (NaI, $LaBr_3$), which emits optical photons collected by PMTs or silicon photomultipliers (SiPMs).
- A gamma ray initiates an electromagnetic cascade in the upper atmosphere (~ 10 km altitude). The shower particles travel faster than the local speed of light in air ($n_{\mathrm{air}} \approx 1 + 3 \times 10^{-4}$), emitting **Cherenkov radiation**.

Source–Messenger Summary

Source	GW	ν	CR	γ
NS–NS merger	$\checkmark\checkmark$	$\sim$	$-$	$\checkmark\checkmark$
NS–BH merger	$\checkmark\checkmark$	$\sim$	$-$	$\sim$
BH–BH merger	$\checkmark\checkmark$	$-$	$-$	$-$
Gamma-ray burst	$\sim$	$\sim$	$\sim$	$\checkmark\checkmark$
AGN	$-$	$\checkmark$	$\checkmark$	$\checkmark\checkmark$
Supernova	$\sim$	$\checkmark\checkmark$	$\checkmark$	$\checkmark$
Pulsar/PWN	$\sim$	$-$	$\sim$	$\checkmark\checkmark$

$\checkmark\checkmark$ = confirmed, $\sim$ = expected/tentative, $-$ = not expected.

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Thank You!

Questions?